

Accepted Manuscript

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PII: S0308-8146(19)30735-6
DOI: <https://doi.org/10.1016/j.foodchem.2019.04.072>
Reference: FOCH 24673

To appear in: *Food Chemistry*

Received Date: 17 September 2018
Revised Date: 17 April 2019
Accepted Date: 19 April 2019



Please cite this article as: de Souza Gois Barbosa, M., Brígida dos Santos Scholz, M., Sorane Good Kitzberger, C., de Toledo Benassi, M., Correlation between the composition of green Arabica coffee beans and the sensory quality of coffee brews, *Food Chemistry* (2019), doi: <https://doi.org/10.1016/j.foodchem.2019.04.072>

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Title

Correlation between the composition of green Arabica coffee beans and the sensory quality of coffee brews

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Abstract

This research aimed to correlate the composition of green Arabica coffee beans with the sensory quality coffee brews. The chemical composition of green Arabica coffee bean (66 samples) from three coffee quality contests was analyzed by near-infrared spectroscopy. Coffee brews with lower quality scores were correlated with high levels of caffeine, protein,

chlorogenic acids and total titratable acidity (TTA) in the green coffee beans. High sucrose/TTA and cafestol/kahweol ratios in the green coffee beans were usually associated with higher scores for the coffee brews. By multivariate analysis techniques, the samples were separated into groups according to production years indicating a strong influence of the environmental conditions on the chemical composition. The profile of the composition of the crude coffee can be indicative of the sensory quality of the coffee brews, relevant information for producers and industry since the green beans are the material used for trading and purchasing coffee.

Keywords: Chlorogenic Acids. Sucrose. Caffeine. Diterpenes. Proteins. Near-infrared Spectroscopy

Chemical compounds studied in this article

Caffeine (PubChem CID: 2519); kahweol (PubChem CID: 114778); cafestol (PubChem CID: 30 108052); sucrose (PubChem CID: 5988).

1. Introduction

In 2018/2019, coffee production in Brazil was 63.4 million 60 kg bags, with 46.3 million of those bags being Arabica coffee and the rest being Robusta coffee (USDA, 2019). In 2018 coffee year, Brazil has exported approximately 35.33 million bags of green Arabica coffee to countries including Germany, the United States and Italy (ICO, 2019).

Arabica coffee is known for producing brews with high cup quality. However, agronomic parameters such as edaphoclimatic conditions, and coffee genetics impact the composition of the green coffee beans and, consequently, the resulting coffee brew. Harvest and post-harvest processes, roasting process and storage of roasted coffee are responsible for the production or degradation of several compounds, as carbohydrates,

acids, caffeine, and lipids, which adds sensory characteristics to the coffee beverage (Kitzberger et al., 2013a; Kitzberger; Scholz; Benassi, 2014; Cheng et al., 2016; Rosa et al., 2016).

Sensory attributes such as acidity, body, and sweetness are important for the quality of the coffee brew (Rosa et al., 2016). In addition to impacting the sweetness, simple carbohydrates, represented mainly by sucrose, glucose, and fructose in coffee, also participate in Maillard reactions during the roasting process, affecting the aroma and color characteristics (Murkovic; Derler, 2006). Proteins also participate in these reactions, which form melanoidins and low-molecular-weight volatile compounds (Montavon et al., 2003). Quinic, citric and malic acids contribute to the acidity of the beverage (Rogers et al., 1999; Koshiro et al., 2015), whereas chlorogenic acids are associated with bitterness and participate in the formation of pigments (Schenker; Rothgeb, 2017). Caffeine, a heat-stable compound, is also associated with bitterness (Rosa et al., 2016; Sunarharuma et al., 2014). Lipids contribute to aroma formation during the roasting process due to decomposition and auto-oxidation reactions and to the brew's body (Koskei et al., 2015). For diterpenes, lipids of the unsaponifiable fraction, the literature reports that higher cafestol/kahweol ratios are associated with better cup quality (Novaes et al., 2015).

In recent years, consumers have been looking for high-quality and healthy beverages, increasing the demand for the production of coffees with high sensory characteristics. Regional, national and international quality contests are carried out in many countries as initiatives to develop and enhance the quality of the coffee and to value the producers. One of the most commonly used protocols in coffee quality contests is that established by the Specialty Coffee Association of America (SCAA). Eleven sensory attributes (fragrance/aroma, uniformity, defects, sweetness, flavor, acidity, body, aftertaste, balance, clean cup and overall) are evaluated to generate a final/global score for the beverage classification (Specialty Coffee Association of America, 2015). The global score

is an individual perception of tasters on the coffee brew quality (Lingle; Menon, 2017).

There are few studies in the literature that relate the composition of the green coffee beans to the coffee cup quality (Franca et al., 2005; Farah et al., 2006a), and they are focused on comparing coffees with high differences in quality (soft, hard, rio, and riado) but do not cover coffees with good grades. As green coffee beans are used for trading and purchasing, information that correlates the composition of green bean with the quality of the roasted coffee brews and the identification of possible quality markers is important to producers and industry.

Among the many techniques available to determine the composition of green coffee, we highlight the near-infrared spectroscopy (NIRS). Its application in the coffee research have been occurring from long time (Avelino et al., 2005) until recently with the use of hyperspectral imaging (HSI), which integrates NIRS with imaging, allow to predict of green coffee composition without the need of grinding (directly on the whole green coffee bean) (Caporaso et al., 2018a). Among the main advantages of this technique is the possibility of analyzing a large number of samples with a minimum level of preparation, the avoiding the use of chemical reagents and the possibility of determination of multiple compounds simultaneously. But, difficulties in the preparation of analytical curves and the cost of the equipment still prevent more application of NIRS. Multivariate analysis techniques, such as principal components analysis, allow the reduction of data from a large data matrix without the loss of significant information can be used to obtain the association between overall quality and chemical composition. This technique allows the association of the variables as the samples and also identifies the relationships between the variables in the principal components' formation.

The sensorial attributes of the coffee beverage are the result of the many reactions between the chemical compounds present in the coffee beans that occur during the

roasting. The quantity and quality of these precursor compounds determine the attributes that the tasters perceive in sensory evaluation.

Thus, this research aimed to determine the chemical composition of Arabica coffees from quality contests and to correlate this information with the quality of the coffee brews. The hypothesis considered is that the composition profile of the green coffee beans obtained by a simple nondestructive technique such as NIRS can provide insights regarding the cup quality of the coffee brews mainly for good sensory quality ones.

2. Materials and methods

2.1 Description of the samples

A total of 66 Arabica coffees from the Coffee Quality Paraná contests, conducted in the years 2012, 2013 and 2015, were selected for the study. The Instituto Agronômico do Paraná (IAPAR), who coordinated the contests, provided the samples. All samples came from a traditional region of coffee cultivation of Paraná state, Brazil.

The coffee was prepared by the coffee producers in their properties by natural (NC) or semi-dry (DC) post-harvest processing. After the drying process, a sample (2-3 kg) of dried coffee was separated and sent to the contests committee. The coffee quality contest team prepared the coffee samples for sensory evaluations.

The removal of the husk and the parchment followed by the removal of defective beans were carried out by the competition organization team. Next, the size of the coffee beans was standardized, passing the beans in coffee sieve # 16.

Before the roasting process, about 200 g of green coffee beans were separated and placed in plastic bottles and stored in a cool and dark place until the chemical determinations.

2.2 Roasting coffee

The green coffee beans (800 g) were roasted in a Probat roaster (Probatino model, Leogap, Curitiba, PR, Brazil) at 200 °C and roasting time of 8-10 min and the roasting endpoints were visually determined. All the samples were roasted until they reached a medium roasting in the scale of SCAA (SCAA, 2015). The roasted coffee beans were evaluated on the same day as the roasting process. In order to perform the sensory evaluation, the roasted coffee beans were ground in a Bunn mill model G3 (Bunn, Springfield, USA).

2.3 Sensory Analyses

In the coffee quality contest, the evaluation of the coffee brews was carried out according to the SCAA protocol. Eleven attributes (fragrance/aroma, uniformity, defects, sweetness, flavor, acidity, body, aftertaste, balance, clean up and overall) were evaluated in each sample.

Depending on the year of the contest, the samples were evaluated by 10 to 12 tasters. All tasters were approved in training conducted by SCAA Q Grade tasters of the Brazilian Association of Special Coffee (BASC). Before each contest, the tasters are again given quick training to standardize the identification and intensity of the attributes provided.

According to the SCAA protocol, 5 cups are prepared for each sample containing ground coffee (7.6 g) and covered with aluminum foil. The tasters evaluate the intensity of the fragrance and record on the evaluation sheet. Then add 150 ml of boiling water over the ground coffee, and after 3 min rest the samples are mixed, and the tasters re-evaluate the fragrance. In 8 min after the addition of water, the floating powder is removed from the cups, and when the coffee reaches 60-65 °C, the tasters evaluate the other attributes, noting their notes on the sheet. For the evaluation of the samples, the tasters use the SCAA card, where the attributes are identified. After assessing all attributes, tasters assign

a global score for each sample. These scores represent the overall quality of the brew and will be used to associate with the chemical composition determined in the green coffee bean.

2.4 Chemical composition analysis

For the NIR analysis, the samples of green coffee beans were ground in a Perten 3600 disk mill (Hägersten, Sweden) after being frozen with liquid nitrogen. The particle size was standardized for 24 mesh (0.5 mm) and stored in -18 °C (Kitzberger et al., 2013a).

2.4.1 NIRS analysis

The chemical composition of green coffee beans was determined by using a near infrared spectrometer (6500 Foss NIRSystems, Silver Spring, MD, USA) equipped with diffuse reflectance detector over a spectral region of 400 to 2500 nm. Approximately 6 g of the ground beans were placed in a rectangular cell and gently compressed with a metal spatula, and the cell was closed with a lid. ISIScan software (Foss, Silver Spring, Maryland-USA) was used to control the spectrometer, to collect the spectra, and to import and analyze the data. The NIR spectra were collected at 2 nm intervals from 1100 nm and 2500 nm. The reflectance at each wavelength was stored as $\log(1/R)$. All calculations were performed on WinISI 4.5 software (Infrasoft International, Port Matilda, PA, USA).

The following chemical compounds: proteins, lipids, caffeine, total chlorogenic acids, total sugars, sucrose, kahweol, cafestol were determined in green coffee samples. The analytical curves for respective compounds were built on the same equipment are described by Scholz et al. (2014a) and Scholz et al. (2014b). The analytical ranges for each compound determination were presents in Table S1. All results were expressed as g 100 g⁻¹ of green beans on a dry basis.

The content of total diterpenes ($\text{g } 100 \text{ g}^{-1}$) was obtained from the sum of the kahweol and cafestol contents. The ratios of sucrose/acid (Suc/TTA) and cafestol/kahweol (C/K) were also calculated.

2.4.2 Other chemical analyses

Total titratable acidity (TTA) was evaluated by titration with 0.1 M NaOH solution using a 1% phenolphthalein solution as an indicator, and the results are expressed in mL NaOH (0.1 mol L^{-1}) 100 g^{-1} of green coffee beans (Mendonça, Pereira and Mendes, 2005).

2.5 Statistical analyses

The multivariate analyses PCA were used to associate the chemical composition and sensory evaluation: mean value of all physical-chemical variables and the overall score were used to perform this association.

The presence of possible structured groups within the samples set was explored with the application of hierarchical cluster analysis (HCA). HCA was performed with the same active variables employed in PCA. The hierarchical tree was obtained by the Ward method as the linkage rule and considering the Euclidian distances as the coefficient of dissimilarity.

Pearson correlation coefficient (r) was used to evaluate the strength and direction of the linear relationship between the chemical composition of the green beans and the global quality score of the coffee brews; a significance level of 0.05 was applied.

For all statistical analyses, Xlstat software (Addinsoft, 2010) was used

3. Results and discussion

3.1 Chemical composition of green coffee beans

The contents of sucrose, total chlorogenic acids, nitrogenous compounds (protein and caffeine), total lipids and diterpenes (kahweol, cafestol), and the sucrose/acid (Suc/TTA) and the cafestol/kahweol (C/K) ratios for the green coffee beans are shown in Table S2, i. Information regarding global scores of each coffee is also included. Table 1 summarizes the mean values for all the parameters studied regarding each year of the contest.

The variability (CV% values) observed in the contents of the different compounds ranged from 5 to 39%; it was probably caused by diversity in coffee variety, edaphoclimatic conditions, and post-harvest process. With the exception of the content of lipids and sucrose, and the ratios Suc/TTA and C/K, significant differences among the compounds comparing the years of the contests were found (Table 1). Similar variations have been reported for coffee genotypes grown at different sites (Kitzberger et al., 2013a; Scholz et al., 2011; Scholz et al., 2018). Caporaso et al. (2018b), studying green coffee beans from different locations and genotypes, reported wide variations (CV from 14.0 to 53.3%) in the content of 50 volatile compounds of roasted coffees among batches of roasting.

The sucrose content in the green coffee beans of all samples ranged between 5.13 and 7.76 g 100 g⁻¹ (CV of 9%) with a mean value of 6.66 g 100 g⁻¹ (Table 1 and TS2). For Arabica coffees from other countries, sucrose contents between 5.87 and 11.11 g 100 g⁻¹ (Ky et al., 2001; Knopp; Bytof; Selmar, 2006; Smrke et al., 2015) were reported. Besides the contribution to the sweetness, sucrose and reducing sugars contribute to the color and aroma formation of the coffee brew (Murkovic; Derler, 2006; Rosa et al., 2016). The sugar profile in the green coffee bean is subject to great variability because it is influenced by genetic characteristics and climate conditions, but mostly by maturation when the reducing sugars content decrease and sucrose content increases (Cheng et al., 2016). In general, higher sugar contents are associated with better cup quality (Sunarharuma; Williams;

Smyth, 2014). Mazzafera (1999) reported lower levels of sucrose for defective beans compared to those of good cup quality coffees.

The total titratable acidity (TTA values) ranged from 193.59 to 409.85 mL NaOH (0.1 mol L⁻¹) 100 g⁻¹ (12% CV) (Table 1 and TS2). Mendonça et al. (2005) reported TTA values between 198.42 and 237.64 mL NaOH (0.1 mol L⁻¹) 100 g⁻¹. Values between 3.69 and 8.11 (CV of 15%) were obtained for the sucrose/acid ratio (Suc/Ac) (Table 1 and TS2) as the similar range (from 6.21 to 8.61) reported in the literature for Brazilian Arabica coffees (Kitzberger et al., 2013a). Higher contents of malic and citric acid contribute to the acidity of the beverage; however, the contents of the acids tend to decrease during the ripening of the coffee bean (Rodrigues et al., 2007; Koshiro et al., 2015).

Franca et al. (2005) compared coffees of different cup qualities and observed that an increase in acidity in the green coffee bean was related to a reduction in the cup quality, which the authors attributed to the presence of both immature and fermented beans in the lower quality coffees. In this way, the suc/ac ratio can be related to the maturity and/or sanity of the green coffee bean; that is, higher ratios correspond to higher contents of mature or unfermented beans, which positively impact the quality of the brew.

The total chlorogenic acids content (CGA) presented great variability (CV 39%) with values ranging from 1.90 to 8.88 g 100 g⁻¹ (Tables 1 and S2). A higher CGA content was reported for Brazilian Arabica coffees (5.07 to 9.34 g 100 g⁻¹) (Farah et al., 2006a; Farah et al., 2006b; Duarte et al., 2010) than for those of other origins (from 3.4 to 4.8 g 100 g⁻¹) (Ky et al., 2001; Smrke et al., 2015). Chlorogenic acids are related to the astringency and bitterness in coffee brews (Farah et al., 2006a; Gloess et al., 2013), and their content decreases with increasing coffee bean maturity (Cheng et al., 2016). Although the literature has often associated high levels of CGA with lower cup quality (Farah et al., 2006a), the high variability in the CGA content in coffee beans with good cup quality (Table S2) suggests that it is possible to obtain coffee brews with good sensory qualities

with various levels of CGA. Zanin et al. (2016) also reported that good cup quality roasted coffees could show wide variation in their chlorogenic acid content. These data suggest that cup quality of the coffee brew is associated not only with the contents of some specific components but also with possible interactions between the compounds, as reported by Borsato et al. (2011) and Scholz et al. (2018).

Comparatively, the contents of nitrogenous compounds were less variable, with CVs of 7% and 8% for the protein and caffeine contents, respectively (Table 1), despite these factors being affected by soil and fertilization conditions (Malta et al., 2003).

Protein contents ranged from 12.08 to 16.36 g 100 g⁻¹ (Table S2), which are lower than those described for Brazilian Arabica coffees (between 13.56 and 18.06 g 100 g⁻¹) (Kitzberger et al., 2013a; Kitzberger et al., 2016). Proteins are involved in Maillard reactions during the roasting process, impacting the color and aroma to the coffee brews (Montavon et al., 2003; Rosa et al., 2016). Franca et al. (2005) reported higher protein contents for higher cup quality green coffee beans than for those corresponding to low quality brews.

The caffeine contents ranged between 1.00 and 1.57 g 100 g⁻¹ (Table S2), similar to the range described for Brazilian coffees (0.91 to 1.52 g 100 g⁻¹) (Farah et al., 2006a; Perrone et al., 2008; Kitzberger et al., 2013a), and those of other origins (between 0.96 and 1.62 g 100 g⁻¹) (Ky et al., 2001; Smrke et al., 2015). Caffeine is formed in immature fruits and accumulates during coffee bean development, and it is highly affected by genotype and environmental conditions (Cheng et al., 2016). Caffeine is associated with bitterness and astringency of the coffee brew (Gloess et al., 2013; Sunarharum et al., 2014; Schenker; Rothgeb, 2017).

Among all the compounds studied, the lipid content had the lowest variability (CV of 5%) with values ranging from 13.00 to 16.24 g 100 g⁻¹ (Table S2) and mean value of 14,75 g 100 g⁻¹ (Table 1). Scholz et al. (2016) also reported low variability in the lipid contents of

several accessions of Arabica coffee from Ethiopia. For Brazilian coffees, a wide range in the lipid contents was reported with values ranging from 11.23 to 16.29 g 100 g⁻¹ (Kitzberger et al., 2013a; Kitzberger et al., 2016). In addition to variations due to genotype, environmental factors also affect lipid content, and higher levels of lipids were associated with higher altitudes and a greater shade (Cheng et al., 2016). The lipid fraction influences aroma formation as it releases volatile compounds such as aldehydes, ketones, and alcohols during the roasting process (Sunarharuma et al., 2014) that impact the cup quality. Franca, Mendonça, and Oliveira (2005) observed higher lipid contents in green coffee beans with better cup quality than in those of low quality brews.

The total diterpene contents varied from 1.12 to 1.84 g 100 g⁻¹ (CV of 19%) (Table 1 and Table S2); however, a broader range is described in the literature for Brazilian Arabica coffees (0.79 to 1.22 g 100 g⁻¹) (Kitzberger et al., 2013a; Kitzberger et al., 2013b). Contents of cafestol from 0.23 to 0.85 g 100 g⁻¹ and those of kahweol from 0.52 to 1.24 g 100 g⁻¹ were observed, and both parameters showed high variability (CVs of 26% and 16%, respectively), especially for cafestol (Table 1 and Table S2). Scholz et al. (2016), studying the composition of several accessions of Arabica coffee from Ethiopia, verified that the highest variability was observed for the diterpene content (CVs of 32 to 35%) due to a genetic factor. Lower contents of cafestol and kahweol have been reported for Brazilian Arabica coffees, and their contents range from 0.22 to 0.61 g of cafestol 100 g⁻¹ and from 0.31 to 1.08 g of kahweol 100 g⁻¹ (Kitzberger et al., 2013a; Kitzberger et al., 2013b; Kitzberger et al., 2016).

Novaes et al. (2015) associated an increase in the cafestol/kahweol (C/K) ratio with coffees with better cup quality. In our research, although we were evaluating coffee beans with good cup quality, considerable variability was observed in the C/K ratio, and values between 0.25 and 1.29 (CV of 35%) were observed (Table 1 and Table S2). Probably this variation was caused by the high variability of cultivars presenting in the coffee contests.

Greater variation can be observed in the literature for Arabica coffees, and C/K values between 0.31 and 1.63 were reported (Kitzberger et al., 2013a; Kitzberger et al., 2013b).

The coffee samples also presented different cup quality (Table S2), but the global scores had low variance (CV of 5%) compared to the variability of the composition profile (Table 1 and Table S2).

3.2 Association between chemical composition and brew quality

Due to their metabolic and physiological origin, many chemical compounds can have correlations with each other. In the present study, protein showed a positive and significant correlation with caffeine ($r = 0.71$), CGA ($r = 0.73$) and TTA ($r = 0.30$), and a negative and significant correlation between protein with cafestol ($r = -0.42$) and total diterpenes ($r = -0.38$) was found (Table S3). Scholz (2018) also observed a positive correlation between caffeine and protein and between caffeine and 5-caffeoylquinic acid (the main representative of chlorogenic acids), and Torres (2014) reported a positive correlation between the caffeine and total phenolic compound contents.

The content of total diterpenes showed a positive and significant correlation with cafestol ($r = 0.58$), kahweol ($r = 0.58$), lipids ($r = 0.31$). TTA showed a significant positive correlation with caffeine ($r = 0.49$), lipids ($r = 0.29$), CGA ($r = 0.33$) and proteins ($r = 0.30$). CGA content showed a significant and negative correlation with total diterpenes ($r = -0.42$) and cafestol (-0.34) and positive correlations with caffeine ($r = 0.45$) and proteins ($r = 0.73$) (Table S3).

In order to analyze all data jointly, a PCA were performed.

The first two components of the PCA accounted for 51% of the total variability (Figure 2) and showed eigenvalues of 3.79 and -2.38 for PC1 and PC2, respectively. The variables protein, caffeine, CGA (PC1 +) and cafestol, C/K, Suc/TTA e GS (PC1-) were the

main formers of PC1. The variables total diterpenes, kahweol and lipids showed a high correlation with PC2 (+) becoming its main formers (Fig. 1).

When samples are projected onto the plane formed by these two components, it is noted that the samples of the lower global score are associated with PC1 (+) and presented a combination of high values of proteins, caffeine, CGA and TTA and low values of C/K and Suc/TTA. On the other hand, samples of the higher overall score have high cafestol value, and a combination of low protein, caffeine, CGA and TTA values (Fig. 1).

The combination of the lipid fraction compounds (lipids, kahweol, and total diterpenes) was responsible for the vertical separation of the samples and is associated with both samples of high (sample C2) and low (sample A2) global scores, with little contribution in separating coffee from high and low quality.

During the roasting, the PC1 (+) forming compounds have undergone many chemical reactions, and many products related to sensory attributes such as bitterness, astringent and acidity are formed, so the high perception of these attributes may contribute to the low global scores of the samples on the right side of Figure 1.

In addition, it was verified that sucrose alone had a little contribution to the separation of the samples, but was more effective when evaluated in relation to the acidity. Therefore, it is possible to conclude that the beverages with higher GS present higher intensity of sweetness, lower acidity and low intensities of bitterness and astringency.

Hierarchical cluster analysis assembled the samples according to their chemical composition and GS and also identified the samples belonging to each group (Table 2, Fig. 2S). The contents of the chemical compounds and global score were different for each of the three groups (Table 2).

In both PCA and HCA, the samples are grouped according to the year of production, suggesting the great influence of the environmental conditions on the chemical composition of the coffee. This fact demonstrates that events, such as environmental

conditions, genetic, or maturation conditions, that alter the composition will have a decisive effect on the beverage quality.

Although the sensorial quality varies from year to year, the chemical composition of high quality coffee can be defined by the profile of chemical compounds as shown in the groups formed by HCA (Table 2). So the information on the content of several compounds of the green bean can provide insights regarding the cup quality of the coffee brews..

In the multivariate analysis, the relationship between chemical compounds and overall quality of the coffee brews mimic the assessment of tasters for the perception of attributes formed during roasting from chemical compounds of green coffee beans. Therefore, for an specific correlation between a chemical compound and a sensory attribute, a quantitative sensory analysis is necessary, as described by Scholz et al. (2018) and Borsato et al. (2011).

4. Conclusions

In general, the cup quality of the coffee brews were associated to a set of chemical compounds of the green coffee beans. By multivariate analysis techniques, the samples were separated into groups according to production years indicating a strong influence of the environmental conditions on the chemical composition. Coffee brews with lower quality scores were correlated with high levels of caffeine, protein chlorogenic acids and TTA in the green coffee beans. High cafestol, sucrose/acid and cafestol/kahweol ratios in the green coffee beans were usually associated with higher quality scores for the coffee brews. Thus, the hypothesis that the profile of the composition of the green coffee beans can be indicative of the sensory quality of the brews was confirmed.

Acknowledgment

The authors acknowledge CNPq, CAPES, and Universidade Estadual de Londrina (PROPPG / Escritório de Apoio ao Pesquisador – Edital REVISE) for the support.

Conflict of interest

The authors declare no conflict of interest.

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Figure 1. Projection of variables and samples* in the plane formed by first two principal components

* Pro: Proteins; Caf: Caffeine; C: Cafestol; Lip: Lipids; CQA: Chlorogenic acids; K: Kahweol; TTA: Total Titratable Acidity; TD: Total diterpenes; Suc: Sucrose; Suc/TTA (sucrose/total titratable acidity ratio); C/K (cafestol/kahweol ratio); GS: Global score. Coffees identification: A, B and C represent 2012, 2013 and 2015 contest, respectively (more information on Table S1). *

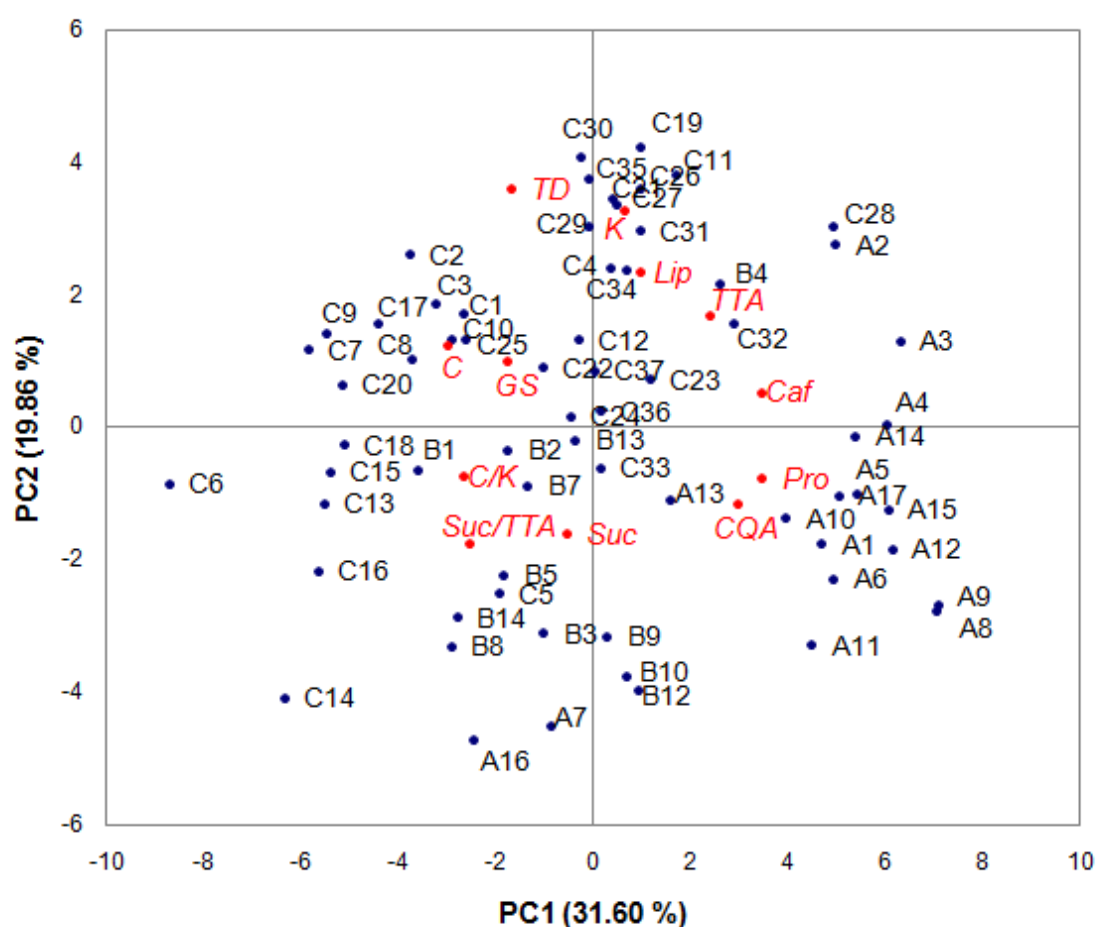


Table 1. Mean values of chemical compounds of green coffee and global score for the coffee brew for each year of contest 2012, 2013 and 2015).

Parameters	Contest			General mean	CV (%)
	2012	2013	2015		
Proteins ¹	15.23 ^a	13.55 ^b	13.63 ^b	14.03	7
Caffeine ¹	1.34 ^a	1.24 ^b	1.25 ^b	1.27	8
Cafestol ¹	0.41 ^b	0.43 ^b	0.58 ^a	0.51	26
Lipids ¹	15.00 ^a	14.43 ^a	14.74 ^a	14.75	5
Chlorogenic acids ¹	8.02 ^a	3.39 ^c	4.03 ^b	4.94	39
Kahweol ¹	0.80 ^b	0.75 ^b	0.92 ^a	0.86	16
TTA ²	325.48 ^a	287.78 ^b	299.77 ^b	304.21	12
Total diterpenes ¹	1.24 ^b	1.24 ^b	1.59 ^a	1.44	14
Sucrose ¹	6.80 ^a	6.80 ^a	6.56 ^a	6.66	9
Cafestol/Kahweol ³	0.54 ^a	0.58 ^a	0.66 ^a	0.61	35
Sucrose/TTA ³	5.28 ^a	5.68 ^a	5.54 ^a	5.50	15
Global score ⁴	77.76 ^b	79.75 ^{ab}	80.52 ^a	79.67	5

Values followed by the same letter in the line do not differ $p < 0.05$

¹g 100 g⁻¹ green coffee bean; ² TTA: Total Titratable acidity - mL NaOH 0,1 mol L⁻¹ 100 g⁻¹ green coffee bean; ³Dimensionless; ⁴Ranging from 0 to 100.

Table 2. Mean values of chemical composition of groups formed in HCA

Samples	Group 1	Group 2	Group 3
	B12, B10, B9, B14, B13, B3, B7, B8, B2, B5, B1, C36, C33, C25, C23, C24, C7, C6, C16, C13, C5, C14, C18, C1, C8, C15, C10, C17, C2, C9, C20, C3, A16, A7	B4, C28, C22, C31, C34, C29, C35, C26, C32, C37, C27, C30, C11, C21, C4, C12, C19	A15, A13, A14, A11, A12, A9, A10, A17, A2, A4, A5, A6, A8, A3, A1
Proteins ¹	13.42 ^c	14.11 ^b	15.31 ^a
Caffeine ¹	1.21 ^b	1.30 ^a	1.36 ^a
Cafestol ¹	0.56 ^a	0.51 ^a	0.39 ^b
Lipids ¹	14.47 ^b	15.04 ^a	15.05 ^a
Chlorogenic acids ¹	3.99 ^b	4.12 ^b	8.04 ^a
Kahweol ¹	0.80 ^b	1.00 ^a	0.83 ^b
TTA ²	287.33 ^b	314.23 ^a	331.13 ^a
Total diterpenes ¹	1.42 ^b	1.63 ^a	1.25 ^c
Sucrose ¹	6.85 ^a	6.20 ^b	6.76 ^a
Cafestol/kahweol ³	0.73 ^a	0.51 ^b	0.48 ^b
Sucrose/TTA ³	5.93 ^a	5.15 ^b	4.95 ^b
Global score ⁴	80.54 ^a	79.69 ^{ab}	77.67 ^b

¹g 100 g⁻¹ green bean; ²mLNaOH 0,1 mol L⁻¹ 100 g⁻¹green bean; ³Dimensionless; ⁴Quality scores ranging from 0 to 100.

Highlights

- Lower brew global scores were linked to high caffeine, protein and CGA contents.
 - High sucrose/acid and cafestol/kahweol ratios in the bean indicate high-quality brew
- High cafestol content in the bean was also associated with high brew global score.